Experiment No. 04

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Batch : SB4

Program:

```
#include <stdio.h>
#include <conio.h>
#define MAX 10
int deque[MAX];
int left = -1, right = -1;
 void input_deque(void);
void output_deque(void);
void insert_left(void);
void insert_right(void);
void delete_left(void);
void delete_right(void);
void display(void);
        int main()
        {
```

```
       int option;
       clrscr();
       printf("\n *****MAIN MENU*****");
       printf("\n 1.Input restricted deque");
       printf("\n 2.Output restricted deque");
       printf("Enter your option: ");
       scanf("%d",&option);
              switch(option)
              {
              case 1:
              input_deque();
              break;
              case 2:
              output_deque();
```

```
              break;
              }
              return 0;
              }
       void input_deque()
       {
       int option;
       do
       {
       printf("\n INPUT RESTRICTED DEQUE");
       printf("\n 1.Insert at right");
       printf("\n 2.Delete from left");
       printf("\n 3.Delete from right");
       printf("\n 4.Display");
```

```
        printf("\n 5.Quit");
        printf("\n Enter your option : ");
        scanf("%d",&option);
                switch(option)
                {
                case 1:
                insert_right();
                break;
                case 2:
                delete_left();
                break;
                case 3:
                delete_right();
                break;
                case 4:
                display();
                break;
                }
                        }while(option!=5);

                }
        void output_deque()
        {
        int option;
        do
        {
        printf("OUTPUT RESTRICTED DEQUE");
        printf("\n 1.Insert at right");
        printf("\n 2.Insert at left");
        printf("\n 3.Delete from left");
        printf("\n 4.Display");
        printf("\n 5.Quit");
        printf("\n Enter your option : ");
        scanf("%d",&option);
        switch(option)
        {
        case 1:
        insert_right();
        break;
        case 2:
        insert_left();
        break;
      case 3:
      delete_left();
      break;
      case 4:
      display();
      break;
      }
              }while(option!=5);
      }
      void insert_right()
      {
      int val;
      printf("\n Enter the value to be added:");
      scanf("%d", &val);
```

```
        if((left == 0 && right == MAX-1) || (left == right+1))
        {
        printf("\n OVERFLOW");
        return;
        }
        if (left == -1)
        {
        left = 0;
        right = 0;
        }
        else
        {
        if(right == MAX-1)
        right = 0;
        else
        right = right+1;
        }
        deque[right] = val ;
        }
        void insert_left()
        {
        int val;
        printf("\n Enter the value to be added:");
        scanf("%d", &val);
        if((left == 0 && right == MAX-1) || (left == right+1))
        {
        printf("\n Overflow");
        return;
        }

        if (left == -1)
        {
        left = 0;
        right = 0;
        }
        else
        {
        if(left == 0)
        left=MAX-1;
        else
        left=left-1;
        }
```

```
deque[left] = val;
}
void delete_left()
{
if (left == -1)
{
printf("\n UNDERFLOW");
return ;
}
printf("\n The deleted element is : %d", deque[left]);
if(left == right)
{
left = -1;
right = -1;
```

```
else
{
if(left == MAX-1)
left = 0;
else
left = left+1;
}
}
void delete_right()
{
if (left == -1)
{
printf("\n UNDERFLOW");
return ;
```

```
printf("\n The element deleted is : %d", deque[right]);
if(left == right)
{
left = -1;
right = -1;
}
else
{
if(right == 0)
right=MAX-1;
else
right=right-1;
}
}
```

```
void display()
{
int front = left, rear = right;
if(front == -1)
{
printf("\n QUEUE IS EMPTY");
return;
}
printf("\n The elements of the queue are : ");

if(front <= rear )
{
while(front <= rear)
{
printf("%d",deque[front]);
front++;
}
}
else
{
while(front <= MAX-1)
{
printf("%d", deque[front]);
front++;
}
front = 0;
while(front <= rear)
{
printf("%d",deque[front]);
front++;
}
}
else
{
while(front <= MAX-1)
{
printf("%d", deque[front]);
front++;
}
front = 0;
while(front <= rear)
{
printf("%d",deque[front]);
front++;
}
}
printf("\n");
}
```

Output :

```
*****MAIN MENU*****
1.Input restricted deque
2.Output restricted dequeEnter your option : 1

INPUT RESTRICTED DEQUE
1.Insert at right
2.Delete from left
3.Delete from right
4.Display
5.Quit
Enter your option : 1

Enter the value to be added:10
```

```
INPUT RESTRICTED DEQUE
1.Insert at right
2.Delete from left
3.Delete from right
4.Display
5.Quit
Enter your option : 1

Enter the value to be added: 20
```

```
5.Quit
Enter your option : 1

Enter the value to be added: 30

INPUT RESTRICTED DEQUE
1.Insert at right
2.Delete from left
3.Delete from right
4.Display
5.Quit
Enter your option : 1

Enter the value to be added: 40
```

```
5.Quit
Enter your option : 4

The elements of the queue are : 10203040

INPUT RESTRICTED DEQUE
1.Insert at right
2.Delete from left
3.Delete from right
4.Display
5.Quit
Enter your option :
```

```
Enter your option : 2

The deleted element is : 10
INPUT RESTRICTED DEQUE
1.Insert at right
2.Delete from left
3.Delete from right
4.Display
5.Quit
```

```
Enter your option : 3

The element deleted is : 40
INPUT RESTRICTED DEQUE
1.Insert at right
2.Delete from left
3.Delete from right
4.Display
5.Quit
```